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Fachbereich Mathematik & Informatik

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Probabilistic Combinatorics SS24

Sheet 6

Due: 27. May at 19:00

(1) For a graph $G = (V, E)$ let $\omega(G)$ the biggest number ℓ, such that G contains a full subgraph K_ℓ. Furthermore let $X_\ell(G)$ be the random variable on $G(n, 1/2)$, which assigns to a graph the number of complete subgraphs K_ℓ of G . Proof the following statements:

(i)
$$\mathbb{E}[X_\ell] = \binom{n}{\ell} 2^{-\binom{\ell}{2}}.$$

(ii) For every $\varepsilon > 0$:
$$\lim_{n\to\infty} \mathbb{P}(\omega \geq (2+\varepsilon)\log n) = 0.$$

(2) In the settings of exercise 1, show the following statements:

(i)
$$\mathrm{Var}(X_\ell) \leq \sum_{k=2}^{\ell} \binom{n}{\ell}\binom{\ell}{k}\binom{n-\ell}{\ell-k} 2^{-2\binom{\ell}{2}+\binom{k}{2}}$$

(ii) For every $\varepsilon > 0$:
$$\lim_{n\to\infty} \mathbb{P}(\omega \geq (2-\varepsilon)\log n) = 1.$$

You may use without proof:
$$\lim_{n\to\infty} \sum_{k=2}^{\ell} \frac{\binom{\ell}{k}\binom{n-\ell}{\ell-k}2^{\binom{k}{2}}}{\binom{n}{\ell}} = 0.$$